

Combining Digital Watermarking and Fingerprinting Techniques to Identify Copyrights

1st Norah Saud Alsubaie
College of Computer Science and
Information Technology
Imam Abdulrahman Bin Faisal
University

2nd Rawan Sudjee AlSaffar
College of Computer Science and
Information Technology
Imam Abdulrahman Bin Faisal
University

3rd Zahra Subhi Al Ubaid
College of Computer Science and
Information Technology
Imam Abdulrahman bin Faisal
University

4th Haifa Abdullah Alhashim
College of Computer Science and
Information Technology
Imam Abdulrahman bin Faisal
University

5th Sarah Hamdan Alqurashi
College of Computer Science and
Information Technology
Imam Abdulrahman bin Faisal
University

Abstract— Multimedia content, which can be easily transferred via online distribution channels, is exposed to risks associated with illegal use. Therefore, security solutions are needed to protect multimedia content from any unauthorized copying and redistribution. The majority of multimedia security systems implement digital watermarking and fingerprinting methods for purposes of content ownership and identification. While digital watermarking is used to hide ownership information into the multimedia file, fingerprinting is used to assign an individual identifier to the multimedia copy in order to identify the case of illegal redistribution. This paper provides a literature review of the integration of digital watermarking and fingerprinting for purposes of copyright protection. Besides, the paper analyses the evolution of robust, fragile and hybrid multimedia security systems, including the use of cryptography, hash functions, authentication tools and collusion-resistance techniques. Also, the types of attacks on multimedia systems are considered, and their influence on the efficiency of the systems is discussed. The current research highlights the challenges associated with the development of multimedia systems, such as scalability problems, the need to ensure imperceptibility, high computational complexity, etc. Finally, this paper identifies the research gaps and needs in developing hybrid multimedia protection systems.

Keywords— Digital Watermarking, Multimedia Fingerprinting, Copyright Protection, Multimedia Security, Robust Watermarking, Traitor Tracing, Cryptography, Collusion Resistance, Authentication, Multimedia Protection

I. INTRODUCTION

The rise in popularity of digital multimedia technologies and online platforms for distributing digital content necessitates efficient copyright protection schemes. The proliferation of multimedia content may result in its unauthorized distribution, modifications, copying, and piracy due to the ease with which any user can share this data. Consequently, there was a need for researchers to design multimedia copyright protection schemes.

The most popular and widely used technique for copyright protection of multimedia content is digital watermarking. This process consists of embedding hidden information into multimedia content in a way that preserves adequate quality of media. The choice of watermarking technique depends on the application's requirements and the

desired level of security. For example, frequency domain watermarks, like those utilizing Discrete Cosine Transform (DCT) or Discrete Wavelet Transform (DWT), offer a higher degree of security as compared to spatial-domain techniques.

Another popular technology that is applied in protecting multimedia content is multimedia fingerprinting. This process aims at tracking and identifying unauthorized redistributors by inserting fingerprints into multimedia content copies. While digital watermarking allows tracking a source of the leak, multimedia fingerprinting traces individual copies that are being disseminated. Fingerprinting schemes incorporate cryptography and collusion-resistance.

In this research, advancements made in digital watermarking and digital fingerprinting technologies to protect copyrights are discussed. Some of the various techniques used in protecting copyrighted material, various attacks on multimedia and the problems faced by current approaches are covered.

II. LITERATURE REVIEW: WATERMARKING AND FINGERPRINTING FOR COPYRIGHT PROTECTION

A. Digital Watermarking Techniques

a. Spatial Domain and Frequency Domain Techniques

Digital watermarking methods can be divided into two main categories: spatial domain and frequency domain watermarking techniques, depending on how the watermark is embedded with the host image. In spatial domain methods, the watermark is simply embedded into the pixel values, with the Least Significant Bit (LSB) method being the most common approach due to its simplicity and low computational cost. However, these methods are vulnerable to common image processing operations and are therefore not as resistant to attacks like compression or noise. Conversely, in frequency domain methods, the watermark is embedded into the transformed coefficients using other techniques like the Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT). This approach improves the robustness of

the watermark since it becomes resistant to manipulations such as compression, filtering, and resizing. In practice, Previous studies indicate that frequency domain methods tend to be preferred in practical applications, as they are more resilient, while spatial domain methods are still useful in applications where simplicity and efficiency are important factors to consider [1].

b. Robust, Fragile, and Semi-Fragile Watermarking

Watermarking techniques are divided into robust, fragile, and semi-fragile methods. Robust watermarking is designed to resist various types of intentional and unintentional changes, such as compression, noise, and geometric transformations, which makes it suitable for copyright protection and ownership verification. In contrast, fragile watermarking is highly sensitive to any changes in the host media, where even minor modifications can damage or remove the watermark. This attribute also makes it especially useful in authentication and integrity checking, as it allows for accurate detection of tampering. Semi-fragile watermarking falls between the two methods, enabling it to tolerate unintentional changes, but still detects intentional alterations. Thus, the type of watermarking is mostly determined by the needs of the application, where robust watermarking is preferred to protect ownership, and weak watermarking is preferred to detect any illegal changes [2].

Recent studies show that watermarking techniques are divided into three types: robust, fragile, and semi-fragile. Robust watermarking can resist attacks, while fragile watermarking is very sensitive and can detect any changes. Semi-fragile watermarking is in between, as it allows normal changes but still detects intentional ones. This grouping shows that there is a trade-off between robustness and sensitivity, which is an important factor when designing modern watermarking systems [3].

c. Imperceptibility and Image Quality Metrics

In digital watermarking, imperceptibility is a key requirement since the watermark embedded in the host image should not introduce noticeable distortion to the image. Image quality is usually evaluated based on imperceptibility, and rather than as a separate parameter. Studies use measures such as Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) to evaluate the quality of a watermarked image. A higher PSNR value shows better image quality with lower distortion, while SSIM checks how similar the watermarked image is to the original image based on their structure. These metrics are widely used to

ensure that the watermark remains imperceptible to the human visual system and at the same time the overall integrity of the media is not compromised. In addition, the literature highlights that there is a trade-off between imperceptibility and other watermarking requirements such as robustness and capacity. Improving robustness or increasing embedding capacity can result in a compromise of image quality and therefore a balance must be established between these competing factors in designing watermarking systems [4].

B. Multimedia Fingerprinting Systems

d. User-Specific Fingerprinting

One important concept in multimedia security is user-specific fingerprinting, in which each distributed copy of digital data is uniquely linked to a specific user. This method allows identifying users based on unique patterns or embedded identifiers and making it possible to trace illegal distribution. In some studies, fingerprinting is derived from user behavior, where individual activity patterns and usage characteristics are used to generate unique identifiers for each user. These behavioral fingerprints allow systems to distinguish between users with high accuracy, as each user exhibits distinctive interaction patterns that can be modeled and recognized [5].

Besides behavior-based methods, some methods create fingerprints using device settings and content features. Use device data, app data, and user preferences to make each user different. Studies show they can identify users or devices with high accuracy, often above 90%, even if some features change. This shows that user-specific fingerprinting is effective in identifying users and linking distributed content to its original source. Overall, both behavior-based and content-based methods of fingerprinting are important in supporting user identification in multimedia systems [6].

e. Traitor Tracing

A key component of multimedia fingerprinting systems is known as traitor tracing and is intended to detect users who illegally redistribute protected content. In these systems, each copy of the content contains a unique identifier linked to a specific user. When an illegal copy is found, the fingerprint can be extracted and used to identify its source. Fingerprints can be recovered and analyzed to trace the source of the leak. The given process allows identifying the malicious users and supports copyright protection. Practical traitor tracing frameworks typically involve three main stages: fingerprint embedding, detection, and tracing, where the extracted identifier is used to determine the origin of the redistributed content [7].

Recent research highlights the importance of fingerprinting methods in addressing piracy and ownership verification issues. Specifically, fingerprint-based techniques have been applied not only to multimedia content, but also to complex systems such as deep neural network models, where unique fingerprints can be used to identify unauthorized use and verify intellectual property ownership. These methods enable accurate distinction between original and pirated models, even when the attackers are trying to modify or reuse the content that is safeguarded. Moreover, fingerprinting methods provide a non-repudiable mechanism for ownership verification, strengthening their role in combating digital piracy and ensuring accountability in content distribution systems [8].

f. Collusion Resistance in Fingerprinting Systems

Collusion resistance is an important requirement in multimedia fingerprinting systems, where multiple users may work together to hide or remove fingerprints to avoid being identified. In these attacks, users combine different copies of the same content to create a new version where the fingerprint becomes weaker or unclear. To address this challenge, advanced fingerprinting systems employ collusion-resistant codes that ensure the persistence of sufficient identifying information even after multiple copies are merged. These codes enable the system to trace and identify one or more colluding users by analyzing the extracted fingerprint from the redistributed content.

Recent approaches integrate collusion-resistant fingerprinting into secure multimedia distribution frameworks to enhance both traceability and security. For instance, modern systems utilize robust fingerprint generation schemes, such as Nuida codes, which are designed to tolerate a predefined number of colluders while maintaining reliable detection performance. In such systems, a tracing algorithm is applied to the extracted fingerprint to identify the most likely colluding users based on correlation or scoring mechanisms. Additionally, these approaches highlight a trade-off between the level of collusion resistance and system performance, where increasing the number of supported colluders typically requires longer codewords, which may impact computational efficiency and content quality. Overall, collusion-resistant fingerprinting plays a key role in ensuring the security and reliability of multimedia distribution systems against cooperative attacks [9].

C. Hybrid Watermarking and Fingerprinting Systems

g. Hybrid Framework Design

Hybrid watermarking and fingerprinting systems have been developed to improve multimedia security by combining the strengths of both techniques into a single system. Watermarking is used in these systems to embed ownership or authentication data, with fingerprinting being used to identify users and trace content distribution. One such representative method is embedding fingerprint data as a watermark, where the system can simultaneously provide both authentication and user identification. This combination offers more comprehensive protection than applying either technique on its own.

Recent research suggests hybrid architectures that use transform-domain methods where multiple transformations are combined to enhance the performance of the system. An example is the hybrid framework that uses lifting wavelet transform (LWT) and discrete wavelet transform (DWT) to achieve better robustness and imperceptibility. In this method, adaptive strategies are used to embed the fingerprint information into the low-frequency band of the host image, without noticeable distortion while remaining resistant to various attacks. The framework also includes the use of feature extraction algorithms, including local binary patterns (LBP) to adjust embedding parameters dynamically, enhancing overall system efficiency. The proposed system, also, is shown to perform well in terms of image quality and robustness, as demonstrated through evaluation metrics like PSNR, SSIM, and normalized correlation coefficient (NCC). These findings support the effectiveness of hybrid watermarking–fingerprinting systems in offering secure, reliable, and high-quality multimedia protection solutions [10].

h. Integration with Cryptography

The combination of cryptography with watermarking and fingerprinting systems plays an important role in improving multimedia security. Although watermarking and fingerprinting provide the means of ownership authentication and user identification, cryptography adds an extra layer of protection in securing the ownership and user identification of the embedded information. In contemporary hybrid systems, encryption is applied during the watermarking process to protect the watermark information as well as the host media against tampering and interception.

Recent methods have proposed encryption-based watermarking schemes that combine a combination of signal processing algorithms and cryptographic operations to provide a higher degree of security. For example, techniques like discrete wavelet transform (DWT) and discrete cosine transform (DCT) are combined with encryption to protect

watermark data during embedding and transmission. These systems ensure that even if the watermark is found, it cannot be understood without the correct key. They also help prevent unauthorized users from extracting or changing the watermark. Overall, combining cryptography with watermarking and fingerprinting creates a stronger, multi-layer security system that protects data and improves the reliability of multimedia applications [11].

i. Robustness Against Attacks

Resistance to attacks is an essential requirement in watermarking systems since the information embedded should still be identifiable after various signal processing operations. The most common types are compression, noise addition, filtering, and geometric transformations, which may cause degradation or removal of the embedded watermark. Among these, recompression attacks represent a major challenge, especially in video distribution systems where the content is often re-encoded under varying compression settings.

Recent studies have been directed towards the development of robust watermarking schemes that can withstand such attacks without compromising on the quality of the media. As an example is that compressed-domain watermarking algorithms have been suggested to specifically address recompression attacks by exploiting invariant characteristics of video content. These approaches utilize texture and motion information to adaptively select embedding regions that remain stable under varying compression conditions. By embedding watermark data into such resilient regions, the system improves its ability to resist distortion caused by changes in quantization parameters. Experimental evaluations demonstrate that these methods achieve strong robustness, as measured by metrics such as bit error rate (BER) and similarity, while maintaining acceptable perceptual quality. This balance between robustness and visual quality highlights the effectiveness of modern watermarking techniques in resisting real-world attacks [12].

J. Performance Evaluation of Watermarking and Fingerprinting Systems

Performance evaluation is important in watermarking and fingerprinting systems because it helps measure robustness, imperceptibility, and accuracy. In general, metrics such as peak signal-to-noise ratio (PSNR), bit error rate (BER), and correlation coefficient (CC) are used to measure image quality, watermark stability, and similarity between the original and extracted data.

In recent research, comprehensive evaluation frameworks have been proposed to analyze watermarking performance for various attack scenarios. For example, evaluation of one document image watermarking system showed that the system maintained both robustness and image quality against various attacks such as StirMark benchmark attacks and paint-based attacks. The findings revealed that the proposed method achieved high PSNR values of around 43 dB, and very low BER values, indicating strong resistance to distortion and minimal degradation of embedded information. In addition, the proposed method was compared with several traditional watermarking methods, and the proposed method achieved better detection accuracy and recovery performance than traditional methods. These results highlight the importance of using multiple evaluation metrics and realistic attack scenarios to ensure the reliability and efficiency of modern watermarking systems [13].

Cryptography in Copyright Protection Hashing

To achieve integrity and authentication for content, hashing techniques would be used in detail with Multimedia copyright protection systems from unauthorized modifications. A hash function is used to convert multimedia data into a fixed-length digest from the original input, where a small change in the original content generates an output value that differs greatly from the original image [11]. This property allows them to verify honestly the authenticity of digital content without revealing the actual media itself. Recent studies indicate that hash-based copyright verification methods are commonly integrated with watermarking systems to strengthen ownership authentication and tamper detection capabilities. By comparing generated hash values with the original stored digest, multimedia systems can identify whether the content has been altered during transmission, storage, or redistribution. In addition, cryptographic hashing improves the reliability of copyright protection frameworks because it provides fast integrity verification while maintaining computational efficiency in multimedia security applications [14].

Encryption

Without encryption, illegal access to multimedia content being played is possible and is why encryption is viewed as a basic building block in the realm of multimedia copyright protection. Most present copyright protection architectures are essentially hybrid frameworks that integrate some type of encryption watermarking systems together with fingerprinting methods to deploy multi-layered multimedia security architectures. In these systems, encryption algorithms secure embedded watermark and fingerprint information during both transmission and storage processes. In recent years, new schemes that combine the transform-domain

techniques (eg., DDwt and DCT) with cryptographic operations are proposed to achieve better confidentiality and protection on watermarked embedded copy-right information. The techniques used to accomplish this make sure that if watermark data is extracted, there is still no information available from it unless you have the relevant decryption key. In addition, there are also many other features of encryption mechanisms in the way of secure content distribution that reduce possibilities for unauthorized interception of messages, watermarking and illegal copying multimedia files [15].

Authentication

Authentication mechanisms are essential in multimedia copyright systems because they verify ownership legitimacy and ensure the integrity of digital content. Watermarking methods are essential for embedding hidden authentication information directly within multimedia files, such as video, image and audio. Authentication applications can really benefit from fragile and semi-fragile watermarking approaches, in this case these kinds of methods are obviously capable to detect unauthorized experiments with the current host media. The vision for secure ownership verification and reliable authentication of multimedia systems has recently been enhanced by integrating cryptographic verification methods within watermarking schemes. In hybrid systems, embedded watermark information is compared with original authentication data or cryptographic signatures to verify whether the multimedia content remains authentic and unaltered. These approaches support secure ownership verification while also improving resistance against forgery and unauthorized content manipulation [16].

Attacks on Existing Systems

Watermark Removal

Watermark removal attacks represent a major threat to multimedia copyright protection systems because attackers attempt to weaken or eliminate embedded watermark information while preserving acceptable media quality. Common watermark removal techniques include filtering, compression, geometric transformations, noise addition, and signal averaging operations. Thus, these attacks target the vulnerabilities of embedding algorithms to minimize watermark detection capability or even eliminate embedded ownership information completely. Spatial-domain watermarking methods are most vulnerable because the watermark data is embedded in the image pixel itself. Although frequency-domain methods provide stronger robustness against processing operations, advanced signal processing attacks can still compromise watermark integrity under severe attack conditions. Recent studies emphasize the importance of developing adaptive and robust watermarking strategies capable of maintaining watermark detectability against various real-world attacks [17].

Collusion Attacks

Collusion attacks pose a significant challenge in multimedia fingerprinting systems because multiple users cooperate to hide or remove identifying fingerprints from redistributed content. In these attacks, several differently fingerprinted copies of the same multimedia file are combined to generate a new version in which individual fingerprints become weakened or partially eliminated. To address this issue, modern fingerprinting systems employ collusion-resistant coding techniques designed to preserve sufficient identifying information even after multiple copies are merged. Research highlights that robust fingerprint generation methods, including collusion-resistant codes, can tolerate a predefined number of colluding users while maintaining reliable tracing performance. However, increasing collusion resistance generally requires longer codewords and higher computational complexity, which may negatively affect system efficiency and scalability [18].

Tampering

Tampering attacks, refer to the unauthorized changes to multimedia content with a view of altering/faking/modifying digital information while maintaining visual fidelity (perception). Image cover-up ways like object insertion, removal/duplication and style changing for misleading copyright verify processing. Fragile and semi-fragile watermarking methods are commonly used for tampering detection because they can identify modifications by analyzing changes in the embedded watermark structure. Recent multimedia forgery detection systems combine watermarking with cryptographic verification and feature extraction algorithms to improve authentication accuracy and tamper localization performance. Experimental findings demonstrate that these techniques effectively detect altered image regions while maintaining acceptable robustness against normal multimedia processing operations. Consequently, tampering detection remains an important research area in multimedia copyright protection systems [19].

Challenges and Limitations

Scalability

Scalability represents a significant challenge in multimedia watermarking and fingerprinting systems, particularly in large-scale multimedia distribution environments involving a high number of users and protected files. As the number of distributed copies increases, fingerprint generation, embedding, storage, and tracing processes require greater computational resources and processing time. In collusion-resistant fingerprinting systems, supporting more users often requires longer fingerprint codewords and more complex tracing algorithms, which increases storage overhead and reduces system efficiency. Recent studies indicate that scalability limitations reduce the practicality of many existing copyright protection systems in

cloud-based multimedia distribution platforms and large digital libraries. Therefore, developing lightweight and computationally efficient protection frameworks remains an important objective for modern multimedia security research [20].

Trade-Offs

It is the challenges of making a trade-off between robustness, imperceptibility and embedding capacity at the same time in digital watermarking systems. A higher robustness of the watermark reduces the attack capacity for compression, filtering, and geometric distortions but stronger embedding is associated with stronger visual distortion which lowers quality in media. Similarly, increasing embedding capacity enables more copyright information to be inserted into multimedia content but may negatively affect both imperceptibility and robustness. Existing literature consistently identifies this relationship as a fundamental trade-off in watermarking system design. Consequently, many modern watermarking techniques utilize adaptive embedding strategies to dynamically balance these competing requirements according to media characteristics and expected attack conditions [21].

System Limitations

Despite considerable advancements in multimedia copyright protection technologies, existing watermarking and fingerprinting systems continue to exhibit several limitations. Most methods that perform well in controlled experimental settings do not provide reliable performance for the practical multimedia distribution use case, which includes repeated compression, transcoding, resampling and cross-platform sharing. Furthermore, due to high computational complexity and the sophisticated synchronization mechanism that are required by some protection systems, this will limit their applicability into real-time multimedia applications [1]. Studies estimate that there is no universal watermarking or fingerprint approach that can generally optimise all five key requirements (robustness, imperceptibility, embedding capacity, scalability and resilience against attack) at the same time. These limitations indicate a need for integrated and adaptive copyright protection mechanisms that can sustain a high level of security while maximizing multimedia quality preservation with low computational cost. [22], [23].

III. GAP ANALYSIS OF WATERMARKING TECHNIQUES

IV. TABLE I. WATERMARKING TECHNIQUES AND TYPES

Reference	Technique Type	Main Contribution	Limitation
[1]	Spatial and Frequency Domain	Compared LSB, DCT, and DWT	Spatial-domain methods

	Watermarking	methods and showed that frequency-domain techniques provide stronger robustness against attacks	remain weak against compression, filtering, and noise
[2]	Robust, Fragile, and Semi-Fragile Watermarking	Explained the role of each watermarking type in copyright protection and tamper detection	Difficulty in balancing robustness and sensitivity in one framework
[3]	Semi-Fragile Watermarking	Highlighted the trade-off between robustness and tamper detection in modern systems	Existing methods still struggle to achieve high robustness with accurate tamper detection

Table I summarizes the major watermarking techniques and classifications used in multimedia security systems. The studies indicate that frequency-domain methods generally provide stronger robustness, while robust, fragile, and semi-fragile watermarking techniques serve different security purposes. However, balancing robustness, sensitivity, and efficiency remains a challenge in modern watermarking systems.

II. TABLE II. IMPERCEPTIBILITY AND PERFORMANCE EVALUATION

Reference	Evaluation Metric	Purpose	Identified Gap
[4]	PSNR and SSIM	Evaluated image quality and imperceptibility of watermarked images	Improving robustness or embedding capacity may reduce image quality
[13]	BER, PSNR, and Correlation Metrics	Measured robustness, detection accuracy, and recovery performance under attacks	Traditional methods show limited performance against complex attack scenarios

Table II presents the common evaluation of metrics used in watermarking and fingerprinting systems. The findings demonstrate that imperceptibility and robustness are critical requirements in multimedia protection. Nevertheless, maintaining high image quality while resisting attacks remains a major research challenge.

III. TABLE III. FINGERPRINTING AND TRAITOR TRACING SYSTEMS

Reference	Fingerprinting Method	Application	Limitation
[5]	Behavior-Based Fingerprinting	Used user interaction patterns for identification and tracking	Behavioral patterns may change over time
[6]	Device and Content-Based	Identified users and devices using	Identification accuracy may decrease when

	Fingerprinting	system configurations and content features	device settings change
[7]	Traitor Tracing Framework	Traced unauthorized redistribution using embedded fingerprints	Tracing efficiency becomes difficult in large-scale systems
[8]	Ownership Verification Fingerprinting	Verified ownership and detected piracy in multimedia and AI models	Limited resistance against advanced modification attacks
[9]	Collusion-Resistant Fingerprinting	Applied collusion-resistant codes to identify cooperating attackers	Higher collusion resistance increases computational complexity

Table III highlights the role of fingerprinting and traitor tracing techniques in multimedia security systems. The studies demonstrate that fingerprinting methods support user identification, piracy detection, and ownership verification. However, maintaining robustness against collusion and adaptive attacks remains challenging.

IV. Table IV. Hybrid and Cryptography-Based Protection Systems

Reference	Protection Technique	Security Contribution	Research Gap
[10]	Hybrid Watermarking–Fingerprinting Framework	Combined watermarking and fingerprinting to improve ownership protection and user tracing	Increased system complexity and computational overhead
[11]	Cryptography-Based Watermarking	Integrated encryption with DWT and DCT to improve multimedia security	Secure key management and integration remain challenging
[12]	Attack-Resistant Watermarking	Improved resistance against recompression and signal-processing attacks	Difficult to maintain robustness without affecting visual quality

Table IV summarizes hybrid and cryptography-based multimedia protection systems. The studies show that integrating watermarking, fingerprinting, and cryptographic techniques improves robustness and security. However, balancing computational efficiency, security, and media quality is still an open research issue.

Reference	Research Area	Identified Gap	Limitation
[14]	Hash-Based Copyright Verification	Existing hashing methods mainly verify integrity but do not provide direct ownership tracing or user identification	Hash values become invalid after even minor acceptable modifications such as compression or

		capabilities when multimedia content is redistributed illegally.	resizing, reducing flexibility in multimedia environments.
[15]	Encryption for Multimedia Security	Current encryption-based copyright systems focus heavily on confidentiality but provide limited support for robust watermark synchronization and real-time multimedia processing.	High computational complexity and key management overhead may reduce efficiency in large-scale multimedia distribution systems.
[16]	Multimedia Authentication Techniques	Existing authentication frameworks successfully detect tampering but still struggle to balance authentication sensitivity with robustness against normal multimedia processing operations.	Fragile authentication methods may incorrectly classify legitimate modifications as attacks, reducing practical reliability.
[17]	Watermark Removal Attacks	Many watermarking systems remain vulnerable to advanced signal processing attacks designed to weaken or eliminate embedded ownership information.	Spatial-domain watermarking methods especially exhibit weak resistance against filtering, compression, and geometric transformations.
[18]	Collusion Attacks in Fingerprinting System	Current collusion-resistant fingerprinting methods improve tracing accuracy but still face challenges when attackers combine multiple copies to reduce fingerprint visibility.	Increasing collusion resistance generally requires longer fingerprint codes, resulting in higher computational and storage overhead.
[19]	Multimedia Forgery and Tampering Detection	Detection Existing tampering detection systems can identify altered regions but often experience reduced accuracy under complex multimedia editing operations.	Detection reliability decreases when forged regions undergo additional compression, filtering, or resizing operations.
[20]	Scalability in Fingerprinting System	Large-scale multimedia distribution environments require efficient tracing systems capable of supporting millions of users simultaneously.	Current traitor tracing frameworks become computationally expensive and less efficient as the number of users increases.
[21]	Robustness–Imperceptibility–Capacity Trade-Off	Existing watermarking systems continue to struggle with balancing robustness, imperceptibility,	Improving one performance factor often negatively affects the others, preventing the achievement of

		<i>and embedding capacity simultaneously.</i>	<i>an optimal universal solution.</i>
[22]	<i>Limitations of Copyright Protection Techniques</i>	<i>Many proposed watermarking and fingerprinting systems perform well under controlled experiments but lack sufficient evaluation under real-world multimedia distribution conditions.</i>	<i>Existing systems often experience reduced reliability during repeated compression, transcoding, and cross-platform sharing.</i>
[23]	Robust Multimedia Watermarking Systems	Watermarking Systems Modern multimedia watermarking techniques provide improved attack resistance but still lack standardized benchmarking and unified evaluation frameworks.	Different systems use inconsistent evaluation methods, making fair comparison and performance validation difficult across studies.

Table V summarizes the main research gaps and limitations in multimedia copyright protection systems. The studies show that watermarking, fingerprinting, and cryptographic techniques improve security and ownership verification. However, challenges related to scalability, attack resistance, and balancing robustness, imperceptibility, and efficiency still remain.

V. DESCRIPTION OF PROPOSED TECHNIQUE

Based on the analysis of the literature reviewed, this paper proposes an integrated hybrid multimedia copyright protection framework that combines digital watermarking, fingerprinting, and cryptographic techniques into a unified, multi-layered security architecture. The proposed technique aims to address the key limitations identified in existing systems, specifically the trade-off between robustness, imperceptibility, and computational efficiency, while ensuring scalability and resistance against both individual and collusion attacks.

The proposed technique employs a dual-layer embedding strategy. In the first layer, a robust ownership watermark is embedded into the multimedia content using frequency-domain transform methods. Specifically, the hybrid approach utilizes Discrete Wavelet Transform (DWT) combined with Discrete Cosine Transform (DCT) to embed ownership information into the low-frequency coefficients of the host media. Embedding in the low-frequency band ensures that the watermark survives common signal processing operations, such as compression, filtering, and resizing, while maintaining imperceptibility as validated by PSNR and SSIM metrics [10], [11].

In the second layer, a unique user-specific fingerprint is generated for each distributed copy. The fingerprint is derived using a collusion-resistant coding scheme, ensuring that even when multiple users cooperate to combine their copies, sufficient identifying information remains to trace at least one colluding user. The fingerprint is embedded as an additional watermark layer using an adaptive embedding strategy that dynamically adjusts

embedding strength based on local image features extracted via Local Binary Pattern (LBP) analysis [9], [10].

To enhance security, the embedded watermark and fingerprint data are encrypted using a symmetric encryption algorithm prior to embedding. This ensures that even if an attacker successfully extracts the embedded data, the information remains unintelligible without the corresponding decryption key. The combination of DWT-DCT transform-domain embedding with cryptographic protection provides a strong multi-layered defense against both passive and active attacks [11], [12].

Furthermore, a hash-based integrity verification module is incorporated into the proposed framework. A cryptographic hash of the original multimedia content is computed and stored securely. During ownership verification, the extracted watermark and the recomputed hash are compared with the stored reference values to confirm both the authenticity and integrity of the content. This module supports reliable tamper detection, particularly against forgery and unauthorized content modification [14].

The proposed technique is designed with scalability in mind, supporting large-scale multimedia distribution environments with millions of users. The fingerprint generation process uses computationally efficient collusion-resistant codes that minimize storage overhead while maintaining tracing accuracy [20]. The overall architecture is summarized in Fig. 1 and further detailed in Section V.

VI. SYSTEM DESIGN / FRAMEWORK EXPLANATION

The proposed hybrid multimedia copyright protection framework consists of five main functional modules: (1) Preprocessing and Feature Extraction, (2) Watermark Generation and Encryption, (3) Fingerprint Generation, (4) Dual-Layer Embedding, and (5) Ownership Verification and Traitor Tracing. Each module is described in detail below.

A. Preprocessing and Feature Extraction

In the preprocessing stage, the input multimedia content undergoes standard normalization and color space conversion. For image-based content, the host image is transformed from the RGB color space to the YCbCr space, and the luminance (Y) channel is selected for watermark embedding, as it carries the most perceptual information and is most sensitive to quality degradation. The host image is then decomposed using a two-level DWT, producing four frequency sub-bands: LL (low-frequency approximation), LH, HL, and HH (detail sub-bands). The LL sub-band is selected as the primary embedding region due to its energy concentration and robustness against common signal processing operations [1], [10].

To enable adaptive embedding, Local Binary Pattern (LBP) feature extraction is applied to the LL sub-band. LBP features capture local texture information and are used to determine the embedding strength at each

embedding location. Regions with higher texture complexity are assigned higher embedding strength, allowing the system to maximize robustness without compromising imperceptibility [10].

B. Watermark Generation and Encryption

The ownership watermark is constructed as a binary logo or text image representing the copyright owner's identity. The watermark is first scrambled using Arnold Transform to increase security by randomizing the spatial distribution of watermark data. The scrambled watermark is then encrypted using the Advanced Encryption Standard (AES) algorithm with a 128-bit key, ensuring that even if an attacker manages to extract the embedded data, the watermark information remains unreadable without the decryption key [11], [15].

The encrypted watermark is subsequently decomposed using DCT, and its coefficients are prepared for embedding into the DWT-transformed host image. The combination of DWT and DCT provides a two-stage transform-domain embedding that improves robustness against recompression and filtering attacks, as demonstrated in previous studies [11], [12].

B. Fingerprint Generation

A unique fingerprint is generated for each distributed copy of the multimedia content. The fingerprint generation process is based on a collusion-resistant coding scheme, specifically adapting the principles of Nuida codes to generate binary codewords of fixed length for each authorized user [9]. Each codeword serves as a unique identifier assigned to one user, enabling the system to trace illegal redistribution back to the original recipient even when multiple users collaborate to weaken or eliminate the fingerprint.

The fingerprint codewords are generated offline and stored securely in a trusted database. During content distribution, the appropriate fingerprint codeword is retrieved for each authorized user and embedded into their copy of the multimedia content [7], [9]. The fingerprint embedding uses the same DWT-DCT framework as the ownership watermark, but targets the mid-frequency DCT coefficients within the LL sub-band to maintain a balance between robustness and imperceptibility.

C. Dual-Layer Embedding Process

The proposed framework employs a dual-layer embedding strategy. In the first layer, the encrypted ownership watermark is embedded into the low-frequency DWT-DCT coefficients of the host image using a multiplicative embedding rule, which improves resistance against amplitude scaling attacks. In the second layer, the user-specific fingerprint is embedded into adjacent mid-frequency coefficients using an additive embedding rule.

The embedding strength for both layers is adaptively controlled using the LBP texture features

extracted during the preprocessing stage. This adaptive strategy ensures that embedding distortion remains below the perceptual threshold for all regions of the host image, maintaining high PSNR and SSIM values [4], [10]. After embedding, the inverse DWT and inverse DCT are applied to reconstruct the watermarked and fingerprinted multimedia content, which is then ready for distribution.

D. Ownership Verification and Traitor Tracing

The verification process consists of two sub-procedures: watermark extraction for ownership verification, and fingerprint extraction for traitor tracing.

For ownership verification, the received multimedia content undergoes DWT-DCT decomposition, and the embedded watermark coefficients are extracted from the low-frequency region. The extracted coefficients are decrypted using the stored AES key, and inverse DCT and inverse Arnold Transform are applied to reconstruct the ownership watermark. The extracted watermark is then compared with the original stored watermark using Normalized Correlation Coefficient (NCC) to confirm ownership authenticity [10], [13].

In addition, a hash-based integrity check is performed by computing the cryptographic hash of the received content and comparing it with the original hash digest stored in the secure copyright database. If the hash values match, the content is confirmed to be unaltered. If a mismatch is detected, the system flags the content as potentially tampered, and the tampered regions are localized using semi-fragile watermarking analysis [14], [16].

For traitor tracing, the fingerprint is extracted from the mid-frequency region of the DWT-DCT decomposition. A collusion-resistance tracing algorithm is applied to the extracted fingerprint, computing correlation scores between the extracted fingerprint and all registered user codewords. The user or set of users with the highest correlation scores are identified as the most likely source of the unauthorized redistribution [7], [9]. This mechanism ensures accountability in content distribution and supports legal action against identified traitors.

E. Performance Evaluation Metrics

The performance of the proposed framework is evaluated using a comprehensive set of metrics. Image quality and imperceptibility are measured using PSNR and SSIM, where the system targets PSNR values above 40 dB and SSIM values above 0.98. Watermark robustness is measured using BER and NCC under a variety of attack scenarios, including JPEG compression, Gaussian noise addition, median filtering, rotation, and cropping. Fingerprint tracing accuracy is evaluated under collusion attacks involving varying numbers of colluders [4], [13]. System scalability is assessed by measuring processing time and storage overhead as a function of the number of registered users [20].

VII. VI. DISCUSSION AND ANALYSIS

Digital watermarking and fingerprinting techniques play an important role in multimedia copyright protection by preventing unauthorized copying, redistribution, and tampering. The reviewed studies show that frequency-domain techniques such as DCT and DWT provide stronger robustness against attacks like compression and filtering compared to spatial-domain methods.

The analysis also shows that different watermarking methods serve different purposes. Robust watermarking is mainly used for ownership protection, while fragile and semi-fragile watermarking are useful for authentication and tamper detection. However, balancing robustness, imperceptibility, and embedding capacity remains one of the main challenges in modern watermarking systems.

Fingerprinting systems improve multimedia security by assigning unique identifiers to distributed content copies, allowing traitor tracing and piracy detection. In addition, collusion-resistant techniques help identify attackers who combine multiple copies to hide fingerprints, although these methods increase computational complexity and storage requirements.

The reviewed literature further highlights that integrating cryptography with watermarking and fingerprinting improves multimedia security through encryption, authentication, and integrity verification. Despite these advancements, existing systems still face challenges related to scalability, computational efficiency, and resistance against advanced attacks, indicating the need for more adaptive and lightweight multimedia protection frameworks.

The proposed hybrid framework addresses these challenges by integrating DWT-DCT transform-domain embedding, AES encryption, collusion-resistant fingerprinting, and hash-based integrity verification into a unified architecture. By employing adaptive embedding guided by LBP texture features, the proposed system achieves a favorable balance between robustness and imperceptibility, targeting PSNR values above 40 dB while maintaining strong resistance against common multimedia attacks [4], [10], [11]. Furthermore, the dual-layer embedding strategy allows simultaneous ownership watermarking and user-specific fingerprinting in a single pass, reducing computational overhead compared to sequential embedding approaches.

VIII. FINDINGS AND FUTURE WORK

The findings of this study show that combining digital watermarking, fingerprinting, and cryptographic techniques improves multimedia copyright protection and ownership verification. Frequency-domain watermarking methods provide stronger robustness against multimedia attacks, while fingerprinting systems support user identification and piracy tracing. In addition, cryptographic techniques such as encryption and hashing improve data confidentiality, authentication, and integrity verification.

The study also found that current multimedia protection systems still face several challenges, including scalability issues, high computational complexity, collusion attacks, and the trade-off between robustness and imperceptibility. Many systems perform effectively in controlled environments but show reduced reliability under real-world multimedia distribution conditions such as repeated compression and transcoding.

Future work should focus on developing adaptive and lightweight multimedia protection frameworks that provide stronger attack resistance while maintaining high media quality and computational efficiency. Future research may also explore the integration of artificial intelligence and machine learning techniques to improve watermark detection, attack identification, and real-time multimedia security performance.

IX. CONCLUSION

In conclusion, digital watermarking and fingerprinting techniques play an important role in multimedia copyright protection by preventing unauthorized copying, redistribution, and tampering. The reviewed studies show that frequency-domain watermarking methods provide stronger robustness against attacks, while fingerprinting systems support user identification and traitor tracing. In addition, integrating cryptographic techniques such as encryption, hashing, and authentication improves multimedia security, ownership verification, and data integrity. However, existing systems still face challenges related to scalability, computational complexity, attack resistance, and balancing robustness with imperceptibility. Therefore, future multimedia protection systems should focus on developing adaptive, lightweight, and efficient frameworks capable of providing strong security while preserving multimedia quality in real-world environments.

REFERENCES

- [1] M. Begum and M. S. Uddin, "Digital Image Watermarking Techniques: A Review," *Information*, vol. 11, no. 2, p. 110, 2020, doi: 10.3390/info11020110.
- [2] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [3] X. Zhao, *Robust and Semi-fragile Watermarking Techniques for Image Content Protection*, University of Surrey, 2011.
- [4] M. Cedillo-Hernandez, A. Cedillo-Hernandez, F. J. Garcia-Ugalde, and J. C. Sanchez-Garcia, "Semi-Fragile Watermarking Scheme for High-Resolution Color Images: Tamper Identification, Ownership Authentication, and Self-Recovery," *Algorithms*, vol. 19, no. 1, p. 28, 2026, doi: 10.3390/a19010028.
- [5] S. Sharma, J. J. Zou, G. Fang, P. Shukla, and W. Cai, "A Review of Image Watermarking for Identity Protection and Verification," *Multimedia Tools and Applications*, vol. 83, pp. 31829–31891, 2024.
- [6] C. Yang, L. Wang, H. Cao, Q. Yuan, and Y. Liu, "User Behavior Fingerprinting With Multi-Item-Sets and Its Application in IPTV Viewer Identification," *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 2667–2682, 2021, doi: 10.1109/TIFS.2021.3055638.
- [7] A. Kurtz, H. Gascon, T. Becker, K. Rieck, and F. Freiling, "Fingerprinting Mobile Devices Using Personalized Configurations,"

- Proceedings on Privacy Enhancing Technologies, vol. 2016, no. 1, pp. 4–19, 2016.
- [7] A. Rehman, Practical Watermarking for Multimedia Traitor Tracing, Ph.D. dissertation, IMT Atlantique, Brest, France, 2024.
 - [8] Y. Zheng, S. Wang, and C.-H. Chang, “A DNN Fingerprint for Non-repudiable Model Ownership Identification and Piracy Detection,” *IEEE Transactions on Information Forensics and Security*, 2022.
 - [9] A. Qureshi and D. Megías, “Blockchain-Based P2P Multimedia Content Distribution Using Collusion-Resistant Fingerprinting,” 11th Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), 2019, pp. 1606–1615.
 - [10] S. P. Vaidya, “Fingerprint-Based Robust Medical Image Watermarking in Hybrid Transform,” *The Visual Computer*, vol. 39, pp. 2245–2260, 2023.
 - [11] N. Hasan, M. S. Islam, W. Chen, M. A. Kabir, and S. Al-Ahmadi, “Encryption Based Image Watermarking Algorithm in 2DWT-DCT Domains,” *Sensors*, vol. 21, no. 16, p. 5540, 2021, doi: 10.3390/s21165540.
 - [12] H. Ding, R. Tao, J. Sun, J. Liu, F. Zhang, X. Jiang, and J. Li, “A Compressed-Domain Robust Video Watermarking Against Recompression Attack,” *IEEE Access*, vol. 9, pp. 35324–35335, 2021.
 - [13] L. Laouamer and O. Tayan, “Performance Evaluation of a Document Image Watermarking Approach With Enhanced Tamper Localization and Recovery,” *IEEE Access*, vol. 6, pp. 26144–26165, 2018, doi: 10.1109/ACCESS.2018.2831599.
 - [14] G. Zhang, “Cryptographic Techniques in Digital Media Security: Current Practices and Future Directions,” *International Journal of Advanced Computer Science and Applications (IJACSA)*, vol. 15, no. 8, pp. 933–940, 2024.
 - [15] D. Kirovski, H. Malvar, and Y. Yacobi, “Multimedia Content Screening Using a Dual Watermarking and Fingerprinting System,” in *Proc. 10th ACM Int. Conf. Multimedia*, 2002, pp. 372–381.
 - [16] B. B. Zhu, M. D. Swanson, and A. H. Tewfik, “When Seeing Isn’t Believing: Current Multimedia Authentication Technologies and Their Applications,” *IEEE Signal Processing Magazine*, vol. 21, no. 2, pp. 40–49, Mar. 2004.
 - [17] M. Steinebach and S. Zmudzinski, “Countermeasure for Collusion Attacks Against Digital Watermarking,” in *Security, Steganography, and Watermarking of Multimedia Contents VIII*, Proc. SPIE, vol. 6072, 2006, pp. 470–479.
 - [18] H. G. Schaathun, “Attack Analysis for He & Wu’s Joint Watermarking/Fingerprinting Scheme,” in *Proc. Int. Workshop on Digital Watermarking*, 2007.
 - [19] M. Mishra and M. C. Adhikary, “Digital Image Tamper Detection Techniques – A Comprehensive Study,” *International Journal of Computer Science and Business Informatics*, vol. 2, no. 1, pp. 1–15, Jun. 2013.
 - [20] B. Chor, A. Fiat, and M. Naor, “Tracing Traitors,” in *Advances in Cryptology — CRYPTO ’94*, Lecture Notes in Computer Science, vol. 839, Springer, 1994, pp. 257–270.
 - [21] “The Trade-Off Triangle: Balancing Imperceptibility, Robustness, and Capacity in Video Watermarking,” *International Journal of Engineering Research & Technology (IJERT)*.
 - [22] S. Ajankar, “A Survey of Watermarking Techniques for Digital Media Protection,” *OEIL Research Journal*, vol. 22, no. 2, pp. 112–115, 2024.
 - [23] “A Survey on Digital Video Watermarking Techniques and Attacks,” *Computer Systems Science and Engineering*, vol. 47, no. 1, pp. 419–438, 2023.